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$$z^2 - 3z + (1 + 3i) = 0$$

$$D = 9 - 4(1 + 3i) = 9 - 4 - 12i = 5 - 12i$$

$$5 - 12i = (x + yi)^2 = x^2 - y^2 + 2xyi$$

$$\begin{cases} 5 = x^2 - y^2 \\ -12 = 2xy \end{cases}$$

$$\Leftrightarrow \begin{cases} 5 = x^2 + (-y^2) \\ -36 = x^2(-y^2) \end{cases}$$

$$\lambda^2 - 5\lambda - \frac{144}{4} = 0$$

$$D = 25 - 4\left(-\frac{144}{4}\right) = 25 + 144 = 169$$

$$\lambda = \frac{5 \pm 13}{2} = 9 \text{ of } -4$$

$$\begin{cases} \lambda_1 = x^2 = 9 \Leftrightarrow x = \pm 3 \\ \lambda_2 = -y^2 = -4 \Leftrightarrow y = \pm 2 \end{cases}$$

$$(x + yi) \in \{-3 + 2i, 3 - 2i\} \text{ want } -12 < 0$$

$$z = \frac{3 \pm (-3 + 2i)}{2} = \frac{2i}{2} \text{ of } \frac{6 - 2i}{2}$$

$$z_1 = i$$

$$z_2 = 3 - i$$

References